Signal Processing

The signals carrying IoT data are highly likely to face numerous obstacles and can be corrupted by significant amount of noise present in the environment. White Gaussian model has commonly been used to quantify the noise. The types of noise which have been found to degrade the quality of IoT signals vary from the impact noise resulting from high frequency interference and instantaneous disturbance on the initialization of large equipment to changing connections around the participating IoT devices. All these kinds of noise negatively influence the multi-device information fusion system.

Such noises should be filtered out and the transmitted signal should be reconstructed back to its actual form to ensure the accuracy and reliability of the transmitted information. Here, accuracy of IoT solutions is measured in terms of the number of packets reporting correct information, deviation between the reported and actual results and the delivery to correct destination timeliness.

Assuming that the received signal is affected by white Additive Gaussian noise (AWGN) which is also *stationary* in nature, the received signal $y(i)$ can be represented as,

$$y(i) = x(i) + \sigma\epsilon(i),\ i = 1,2,3 \ldots n-1$$

where $x(i)$ is the noise-free transmitted signal, $\epsilon(i)$ representing independent normally distributed random variable and σ representing the intensity of the noise affecting $y(i)$.

Reconstruction of the original signal $x(i)$ from the instantaneous set of $y(i)$ values without actual assuming a specific model for $x(i)$ or $y(i)$ is the primary aim of the process of '*Denoising*'. The most common approach is to recognize noise components as the high frequency components present in the corrupted received signal, apply Fourier transform and then filter out the high frequency components.

Therefore, the most traditional way of denoising signals is based on Fourier analysis and Fourier transform.

Wavelet Transform (WT)

Wavelet Transform (WT) is another method used in denoising signals. On contrary to the traditional Fourier transform, WT is particularly suitable for application of *non-stationary* signals which may instantaneously vary in time. Primarily, the received signal is divided into different frequency components using wavelets. The basis function of WT is scaled based on frequency and a subset of small waves (known as mother wavelet) is used for implementing WT. The mother wavelet is a time-varying window function used for decomposition of $x(i)$ into weighted sets of

scaled versions of *y(i)*. Consequently, using wavelet transform in signal processing is the process of the partial transformation of the spatial domain and the frequency domain, in order to get useful information accurately from it though corrupted with noise.

Since different frequency levels are used for WT, it is quite convenient for analyzing the signal characteristics at different frequencies and detecting removing corrupting noise. Broadly, there are two types of WT, Continuous WT (CWT) and Discrete WT (DWT).

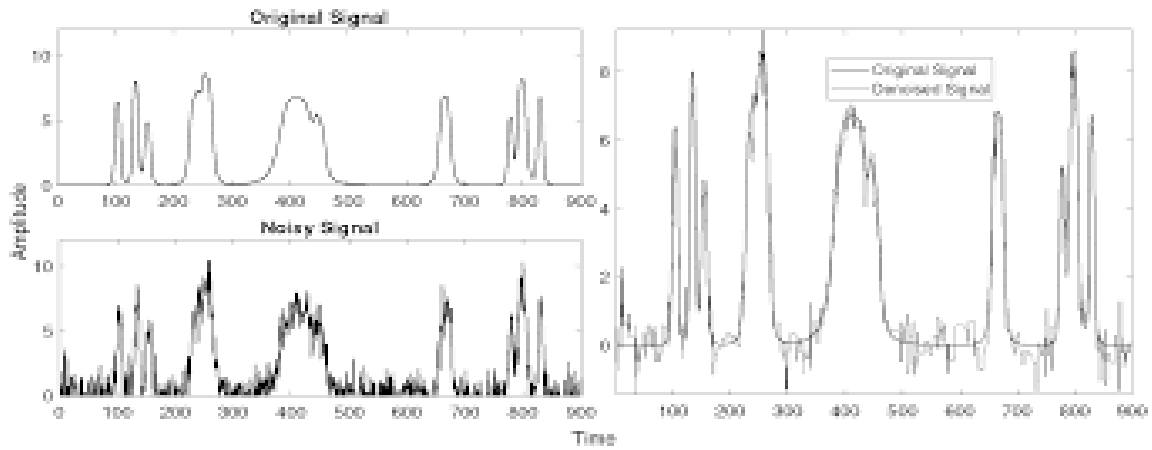
Discrete Wavelet Transform for denoising data

The DWT denoising procedure consists of three steps. In the first step, if the length of the data stream is of length of the order of power of two, it is transformed to the wavelet domain. In the second step, coefficients with either zero magnitude or criterion-based minimized values are selected. In the third or final step, the minimized coefficients are reverted back to the original domain from the wavelet domain to extract the denoised data. DWT-based denoising techniques can be broadly classified into two categories - linear and non-linear.

In linear DWT, signal and noise are assumed to be belonging to the smooth and the detailed part of the wavelet domain, where high frequency components are attenuated. While in nonlinear DWT, the filter removes the coefficients selected in the second stage with amplitudes less than the threshold. In practicality, non-linear DWT is always preferred over linear DWT, as linear DWT introduces error due to the retention of noise components and loss of signal components owing to wavelet filtering.

Whether linear or non-linear DWT denoising technique is used, performance depends on the choice of the wavelet family and the length of the filter. The traditional way for making this choice is based on visual inspection of the data, for example, daublets are implemented when the data appears smooth in the wavelet domain, while Haar or other wavelets are used when the data appears bursty and discontinuous in the wavelet domain.

In order to overcome the problems with DWT denoising, correlation denoising method was introduced. Correlation denoising method implements wavelet transformation and filtering in a way such that the correlation between wavelet coefficients of the signal part and the noise-part is different at each level. However, correlation denoising in its original form is computationally complex. In order to reduce computational complexity, wavelet threshold denoising method was proposed by D. L. Donoho. The method is simple to calculate and the noise can be suppressed to a large extent. At the same time, singular information of the original signal can be preferred well, so it is a simple and effective method. A brief overview of what happens when DWT is applied for denoising is demonstrated in the diagram below.



The four major components of the DWT denoising technique are: wavelet-type selection, threshold selection, threshold function selection and threshold application
to the wavelet coefficients.

1.**Wavelet Selection** - There is a wide variety of wavelets that can be used for denoising. Selecting the optimum one depends on the selection of the matching wavelet filter. Out of different wavelet transform based denoising methods, only minimum description length (MDL) method has the flexibility of choosing the filter type.

2.**Threshold Selection** - There are four basic types of threshold selection, minimax, Stein's unbiased estimate of risk (SURE), and minimum description length (MDL). The ***Universal*** threshold is computed using,

$$t = \sigma \text{ X } \sqrt{(2 \text{ X } \ln(N))}$$

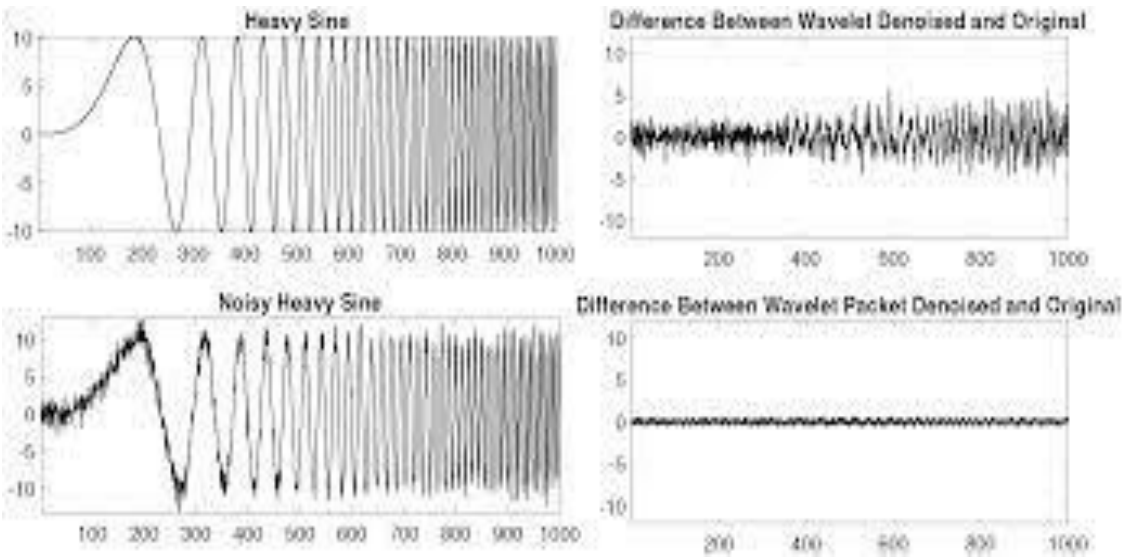
for which N is the length of the signal data array, and σ is the standard deviation of noise. In practicality, in most cases, σ is unknown, but can be estimated using the first detailed part of the wavelet coefficient x_i through the expression,

$$\sigma_{estimate} \approx \frac{\text{Meridian}(|x|)}{0.6745}$$

Signal denoising for IoT networks

The huge amount of sensor data generated in an IoT network are used to take decisions on a certain observation/ phenomenon based on real-time processing. The decision-making procedure often involves detecting the signal energy level transmitted from the sensors. If the received energy level is higher than a predefined threshold, the target phenomenon is detected to be present and vice-versa. However, the sensor data gets crippled with noise contributed by the wireless environment

and the internal electronics of the sensors, on its way to the data center for processing. The WPT method will be the best option in this case for denoising the sensor data, where the original signal coefficients are preserved while removing the noise within the signal. The WPT method can decompose a signal in both scale and wavelet space thereby revealing more details about both the sensor signals and the crippling noise. If energy correlation analysis is used in conjunction with WPT, signal energy from the sensor data can be analyzed and noise can be eliminated by zooming into the signal characteristics at different time scales. Advantages of WPT over WT is evident in the figure below**.** Hence, in this section, a universal framework is presented for denoising sensor signals in IoT networks. The framework is based on energy correlation analysis and combines the processes of WP decomposition, coefficient modification and WP reconstruction.



A key aspect of IoT is event-driven or a periodic sampling. Traditional digital signal processing and control assume periodic samples resulting in time-series data. However, time series consume too much power at the nodes and too much bandwidth on the network. Not all applications are amenable to aperiodic data acquisition.

Constraints on power and bandwidth also encourage distributed computing over sensor events. Relatively small processors can perform useful processing on many data streams. Recognizing interesting events using edge processing reduces the amount of network bandwidth consumed; it also reduces power consumption since wireless communication requires large amounts of power. Cloud computing (centralized servers) or fog computing (servers closer to the edge) can be used to perform further processing on those extracted events.